

The empirical recurrence for OEIS A235243: recovering a conjecture that is not written down

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Abstract

OEIS A235243 records “Empirical recurrence of order 61”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 2744$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 61, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 61 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235243 is “Number of $(n+1) \times (3+1)$ 0..7 arrays with every 2×2 subblock having its diagonal sum differing from its antidiagonal sum by 6, with no adjacent elements equal (constant-stress tilted 1×1 tilings).”. It has offset 1 and begins

3746, 10866, 31858, 108374, 359962, 1305758, 4596034, 17276982, ...

The entry states:

Empirical recurrence of order 61 (see link above).

As of the “Last modified” line on the live entry (Jun 18 2022 23:55:31, revision 8) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 2744$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 2744$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 5488$ exact terms of the model returns order 61 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{61} c_i a(n-i),$$

c_1	13	c_{17}	1728252864706	c_{33}	−6255700947799793367	c_{49}	−3692528116379829696
c_2	96	c_{18}	−48694976871736	c_{34}	−3581456100348940840	c_{50}	−885672213341465856
c_3	−1927	c_{19}	11084154062010	c_{35}	13665502718744553957	c_{51}	995583819668871552
c_4	−1914	c_{20}	434985026528848	c_{36}	4992601559661045450	c_{52}	302300873299247616
c_5	130995	c_{21}	−311181629407490	c_{37}	−23695997476643251017	c_{53}	−182392831539864576
c_6	−177441	c_{22}	−3103066347401222	c_{38}	−5456912202123817121	c_{54}	−69796958576709888
c_7	−5398976	c_{23}	3479746628435524	c_{39}	32530613394490785580	c_{55}	20284519433319936
c_8	15046504	c_{24}	17745742110880816	c_{40}	4924712058580395712	c_{56}	10271694683086848
c_9	149942748	c_{25}	−26338422454159984	c_{41}	−35211142425322385572	c_{57}	−973831565426688
c_{10}	−599186216	c_{26}	−81368113445184984	c_{42}	−4120085348298401408	c_{58}	−861153950601216
c_{11}	−2936111016	c_{27}	149247202228991992	c_{43}	29858028235453286808	c_{59}	−28030449033216
c_{12}	15695327220	c_{28}	298248738086761260	c_{44}	3499549197380780480	c_{60}	30962623610880
c_{13}	40758673440	c_{29}	−656389541919827736	c_{45}	−19635653117346810736	c_{61}	3642661601280
c_{14}	−298504337020	c_{30}	−868881986416189404	c_{46}	−2803316166965940064		
c_{15}	−379570879180	c_{31}	2275269561509800508	c_{47}	9859460290993052768		
c_{16}	4316665341518	c_{32}	1995695588211535835	c_{48}	1829876892289444608		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 61, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $61 + 4$ terms removed returns order 61 as well, so $\deg q_{\min} = 61 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $61 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 61 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A235243.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.